HARSH RAJ

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Research Summary

I work on the evaluation and reliability of AI agents and on open pretraining data for large language models. My work includes Terminal-Bench, a widely used benchmark for command-line agents, and MixtureVitae, an openly licensed pretraining dataset, and has been cited more than 1,000 times.

Education

Northeastern University, Boston, MA — Dec 2026 (expected)
Master of Science in Computer Science — GPA: 3.8/4.0

Delhi Technological University, Delhi, India — August 2023
Bachelor of Technology in Engineering Physics — GPA: 3.4/4.0

Research Experience

Machine Learning Research Intern, Scale AI, New York, NY — May 2026 – Sept 2026
- Studied why AI agents fail on long-horizon tasks. Built a taxonomy that separates errors made by the model from errors caused by the agent harness, and framed root-cause attribution of failures as a search problem.
- *Impact:* Two first-author papers on agent failure taxonomy and root-cause attribution.

Research Engineer Intern, Bespoke Labs, Mountain View, CA — Jan 2026 – Apr 2026
- Built synthetic reinforcement learning environments for training command-line agents. Delivered the environments to frontier AI labs.
- *Impact:* Environments were used to train production CLI coding agents.

Research Assistant, Forschungszentrum Jülich, Jülich, Germany (Remote) — July 2025 – Present
- Led MixtureVitae, building permissively licensed pretraining data for the EU foundation model projects MINERVA, OpenEuroLLM and ELLIOT on the JUPITER supercomputer. Built pipelines for mixing, upsampling, interleaving and quality validation at the scale of hundreds of billions of tokens.
- *Impact:* 10% gain over prior permissive datasets; published in TMLR with a Featured Certification.

Research Assistant, Bau Lab, Northeastern University, Boston, MA — Mar 2025 – Dec 2025
- Studied how pretraining data shapes model behavior and alignment, advised by David Bau and Robert West.
- *Impact:* Co-authored *Synthetic Persona Pretraining*, NeurIPS 2026 AI4Good Workshop.

Research Fellow, AI Risk and Vulnerability Alliance, Remote — Oct 2024 – June 2025
- Contributed bias benchmarks to the UK AI Security Institute's inspect-ai evaluation framework.
- Integrated inspect-ai with the AI Vulnerability Database.

Applied Scientist, Vijil AI, Remote — Jan 2024 – Nov 2024
- Developed vijil-fuzzer, an LLM red-teaming framework with an 85% attack success rate against GPT-4o at launch. Integrated 15+ benchmark datasets into the vijil-evaluate platform and added guardrails to vijil-dome with under 10 ms latency overhead.
- *Impact:* The red-teaming platform is used by enterprise clients including DigitalOcean and Groq.

Publications

* denotes equal contribution; † denotes core team.

Conference and Journal Papers

- **Harsh Raj**, Niranjan Orkat, Suvrorup Mukherjee, Aritra Guha, Cheryl Flynn, Subhabrata Majumdar. Consistency as a Testable Property: Statistical Methods to Evaluate AI Agent Reliability. *NeurIPS 2026.*
- **Harsh Raj**, Ali Elganzory, Marianna Nezhurina, Victor May, Van Khue Nguyen, David Salinas, Huu Nguyen, Jenia Jitsev. Strong Post-Training from Permissive, Reasoning-Dominant, Web-Scale Pretraining. *NeurIPS 2026.*
- Lin Shi, Haowei Lin, Zixuan Zhu, . . . , **Harsh Raj**, . . . , Mike Merrill, Ludwig Schmidt, Alex Shaw. Harbor Adapters and Harbor-Index: Infrastructure and a Curated Meta-Dataset for Large-Scale Agentic Evaluation. *NeurIPS 2026.*
- Mike A. Merrill†, Alexander G. Shaw†, Nicholas Carlini†, Boxuan Li†, **Harsh Raj**†, Ivan Bercovich, Lin Shi, et al. Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces. *ICLR 2026.*
- Huu Nguyen*, Victor May*, **Harsh Raj***, Marianna Nezhurina, Yishan Wang, . . . , Jenia Jitsev. MixtureVitae: Open Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning Data Built from Permissive-First Text Sources. *TMLR 2026.* Featured Certification.

- **Harsh Raj**, Vipul Gupta, Domenic Rosati, Subhabrata Majumdar. Improving Consistency in Large Language Models through Chain of Guidance. *TMLR 2025.*
- Manu Joseph, **Harsh Raj**. GANDALF: Gated Adaptive Network for Deep Automated Learning of Features for Tabular Data. *ACM IKDD CODS-COMAD 2025.*
- Laura F. Nern, **Harsh Raj**, Maurice Georgi, Yash Sharma. On Transfer of Adversarial Robustness from Pretraining to Downstream Tasks. *NeurIPS 2023.*
- **Harsh Raj**, Aakansha Gupta, Rahul Katarya. Extract It! Product Category Extraction by Transfer Learning. *CICT 2021.*

Workshop Papers

- Julian Minder, Viktor Moskvoretskii, Raghav Singhal, . . . , **Harsh Raj**, Huu Nguyen, Jannik Brinkmann, Ashton Anderson, Roland Aydin, Robert West. Synthetic Persona Pretraining: Alignment from Token Zero. *NeurIPS 2026 AI4Good Workshop.*
- Domenic Rosati, Giles Edkins, **Harsh Raj**, David Atanasov, Subhabrata Majumdar, Janarthanan Rajendran, Frank Rudzicz, Hassan Sajjad. Defending Against Reverse Preference Attacks is Difficult. *AAAI 2025 Workshop on AI for Cyber Security.*
- **Harsh Raj**, Domenic Rosati, Subhabrata Majumdar. Measuring Reliability of Large Language Models through Semantic Consistency. *NeurIPS 2022 ML Safety Workshop.* Best Paper Award.
- Sinchani Chakraborty, **Harsh Raj**, Srishti Gureja, Tanmay Jain, Atif Hassan, Sayantan Basu. Evaluating the Robustness of Biomedical Concept Normalization. *NeurIPS 2022 Workshop on Transfer Learning for NLP.*

Preprints and Technical Reports

- **Harsh Raj**, David Lee, Anas Mahmoud, Renxiong Wang, Razvan-Gabriel Dumitru, Chenguang Wang, Tong Zhao, Yunzhong He, Darvin Yi, Vipul Gupta. Root-Cause Attribution Is a Search Problem: Continual Search for Long-Horizon Agent Failures. 2026.
- **Harsh Raj**, Vipul Gupta, Anas Mahmoud, Razvan-Gabriel Dumitru, Darvin Yi, Aakash Sabharwal, Yunzhong He. Model or Harness? An Interaction-Centric Taxonomy for Localizing Agent Failures. 2026.
- Negin Raoof, Richard Zhuang, Marianna Nezhurina, Etash Guha, . . . , **Harsh Raj**, et al. OpenThoughts-Agent: Data Recipes for Agentic Models. 2026.
- Harbor Framework Team. Harbor: A Framework for Evaluating and Optimizing Agents and Models in Container Environments. 2026.
- Manu Joseph, **Harsh Raj**, Anubhav Yadav, Aaryamann Sharma. AskYourDB: An End-to-End System for Querying and Visualizing Relational Databases Using Natural Language. 2022.

Grants, Awards and Honors

- TMLR Featured Certification and Journal-to-Conference Certification for MixtureVitae. 2026
- Swiss AI Large Grant, lead applicant: 200,000 GPU hours on the Alps supercomputer and CHF 61,000. 2025
- EuroHPC Access Grant: 4 million GPU hours on the LEONARDO supercomputer. 2024
- Best Paper Award, ML Safety Workshop at NeurIPS 2022, with a $5,000 prize. 2022
- Rank 7, Codeforces Raif ML Round 1. 2021
- 3rd place, R2R Vision-and-Language Navigation leaderboard at submission, team MLR_Lab_DTU. 2024

Professional Service

Reviewer for ICLR 2027, NeurIPS 2026 Datasets and Benchmarks Track, ACL Rolling Review 2026, EvalEval Workshop at ACL 2026, and ICLR 2026.

Media Coverage and Writing

- Harbor and Harbor-Index used for agent evaluation in LangChain's *How We Benchmark Deep Agents*. 2026
- Harbor-Index covered by Capital & Compute in *Harbor-Index: The AI Benchmark Where Nothing Tops 30%*. 2026
- *Model or Harness?* featured by DAIR.AI and alphaXiv. 2026
- Terminal-Bench reported in Anthropic's Claude launches and tracked by Artificial Analysis. 2025–2026
- *Interpreting the Effects of Jailbreak Prompts in LLMs*, LessWrong. 2024

Open-Source Software

- terminal-bench, benchmark for AI agents in command-line environments; core team, 2,500+ GitHub stars.
- terminal-bench 4, the next version of the Terminal-Bench benchmark; reviewer, 900+ GitHub stars.
- harbor, framework for running and evaluating agents in sandboxed containers; core team, 6,000+ GitHub stars.
- inspect-ai, the UK AI Security Institute's LLM evaluation framework; contributor, 3,000+ GitHub stars.
- garak, NVIDIA's LLM vulnerability scanner; contributor, 5,000+ GitHub stars.